REINDEXING A FINITE AFFINE STRATIFICATION TO THE ORDERED SET OF INTEGERS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 21476-21504, immediately following the statement at 21469-21474.

Editorial note. Four cross-chapter pointers (Morphisms, Schemes) plus two same-chapter lemmas are recorded in assembly.external_references.

Editorial note. The induction on the length of the index set is displayed inside the span.

Allowed standard prerequisites (editorial).

- Affineness of a morphism is insensitive to closing the target up. Morphisms, lemmas morphisms-lemma-closed-immersion-affine, morphisms-lemma-composition-affine and morphisms-lemma-affine-permanence, cited at proof lines 21494-21497. Shows that a morphism T -> Z built from the maximal stratum is affine exactly when T -> X is.
- Affine stratifications and their partial orderings. the definition used in the statement (closure of X_i inside the union of the j <= i strata). Supplies the partial ordering on the index set that the induction consumes.
- Thickenings of a scheme and of a morphism. chapter lemmas lemma-thicken-property-morphisms and lemma-thickening-affine-scheme, cited in the proof. Support the passage from the stratification to the closed subscheme Z with its induced reduced structure.

Source reference. Original lemma 09ZV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 270–270. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06AD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 157–157. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01SG: focused-seeds/geometry/S9-morph-improve-stratification/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01SE: focused-seeds/geometry/S9-morph-improve-stratification/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 01SC: focused-seeds/geometry/S9-morph-improve-stratification/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01I5: focused-seeds/geometry/S9-morph-improve-stratification/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $X = \coprod_{i \in I} X_i$ be a finite affine stratification. There exists an affine stratification with index set $\{0, \ldots, n\}$ where n is the length of I.*

Proof. Recall that we have a partial ordering on I such that the closure of X_i is contained in $\bigcup_{j \leq i} X_j$ for all $i \in I$. Let $I' \subset I$ be the set of maximal indices of I. If $i \in I'$, then X_i is open in X because the union of the closures of the other strata is the complement of X_i. Let $U = \bigcup_{i \in I'} X_i$ viewed as an open subscheme of X so that $U_{red} = \coprod_{i \in I'} X_i$ as schemes. Then U is an affine scheme by Schemes, Lemma 01I5 and Lemma 06AD. The morphism $U \to X$ is affine as each $X_i \to X$, $i \in I'$ is affine by the same reasoning using Lemma 09ZV. The complement $Z = X \setminus U$ endowed with the reduced induced scheme structure has the affine stratification $Z = \bigcup_{i \in I \setminus I'} X_i$. Here we use that a morphism of schemes $T \to Z$ is affine if and only if the composition $T \to X$ is affine; this follows from Morphisms, Lemmas 01SE, 01SC, and 01SG. Observe that the partially ordered set $I \setminus I'$ has length exactly one less than the length of I. Hence by induction we find that Z has an affine stratification $Z = Z_0 \amalg \ldots \amalg Z_{n-1}$ with index set $\{1, \ldots, n\}$. Setting $Z_n = U$ we obtain the desired stratification of X. $\square$